Assignment 5: Multilayer Perceptron with Backpropagation

Group D — Loss Function Comparison

Engin S. Demirkan Franziska Axmann

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1 Introduction

The goal of this assignment is to implement a Multilayer Perceptron (MLP) from scratch using NumPy and evaluate the impact of different loss functions on the MNIST dataset. The loss functions evaluated are Mean Squared Error (MSE), Mean Absolute Error (MAE), and Cross-Entropy (CE).

2 Implemented Loss Functions

For a network predicting y_p with true one-hot labels y_t, the following losses and their gradients w.r.t y_p were implemented.

2.1 Mean Squared Error (MSE)

$$L_{MSE} = \frac{1}{n}\sum(y_p - y_t)^2, \quad \frac{\partial L}{\partial y_p} = \frac{2}{n}(y_p - y_t)$$

2.2 Mean Absolute Error (MAE)

$$L_{MAE} = \frac{1}{n}\sum|y_p - y_t|, \quad \frac{\partial L}{\partial y_p} = \frac{1}{n}\text{sign}(y_p - y_t)$$

2.3 Cross-Entropy (CE)

To ensure numerical stability, the softmax function is computed inside the CE loss. The derivative is streamlined for efficiency.

$$L_{CE} = -\frac{1}{n}\sum y_t \log(\text{softmax}(y_p)), \quad \frac{\partial L}{\partial y_p} = \frac{1}{n}(\text{softmax}(y_p) - y_t)$$

3 Experiments and Results

The experiments were conducted using an MLP with ReLU activations and Stochastic Gradient Descent (SGD) with a learning rate of 0.05 over 20 epochs. A subset of the MNIST dataset was used to accelerate training.

3.1 Loss Curve Comparison

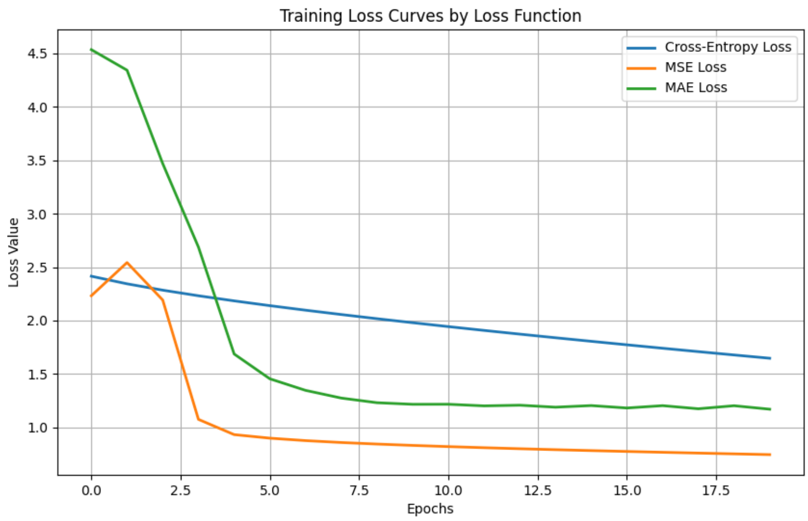

Figure 1: Training loss curves for CE, MSE, and MAE over 20 epochs.

As observed in Figure 1, the Cross-Entropy loss demonstrates a smooth and stable linear decrease. In contrast, MAE and MSE show a sharp initial drop but then plateau quickly. While their absolute loss values appear lower, MSE and MAE suffer from scaling issues in classification tasks and often fail to penalize confident incorrect predictions effectively compared to CE. Cross-Entropy is explicitly designed for probability distributions, making its learning curve the most reliable representation of accuracy improvement.

3.2 Architecture Comparison

We compared a Shallow network (1 hidden layer: 32 neurons) and a Deep network (3 hidden layers: 128, 64, 32 neurons) using Cross-Entropy loss.

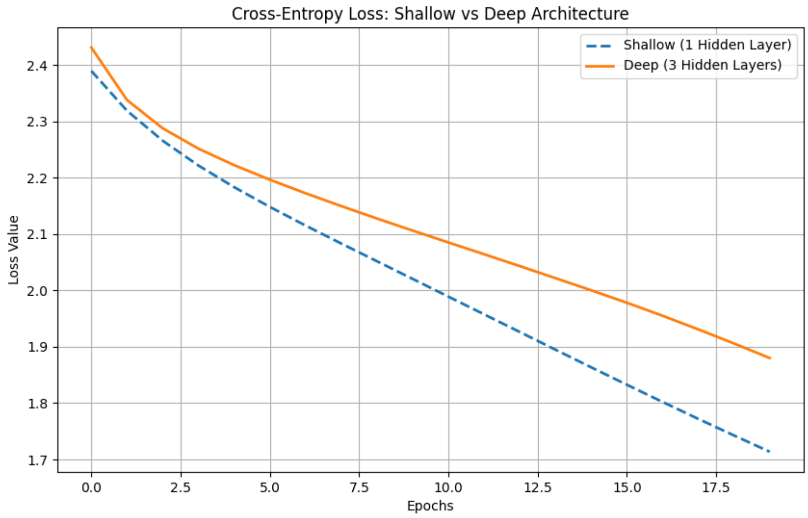

Figure 2: Training loss comparison between Shallow and Deep architectures.

Figure 2 illustrates that the Shallow architecture converged slightly faster and achieved a lower loss within the initial 20 epochs compared to the Deep architecture. This behavior is likely due to the deeper network requiring more epochs to propagate gradients effectively through multiple layers without advanced techniques like Batch Normalization or adaptive optimizers.

3.3 Training Stability and MAE

MAE training is inherently less stable for classification. Because the derivative of MAE relies on the sign function, the gradient magnitude remains constant regardless of how close the prediction is to the target. This prevents the optimizer from taking smaller steps as it approaches a minimum, causing the model to oscillate around the optimal weights rather than converging smoothly.

4 Conclusions

- **Cross-Entropy** is the most appropriate loss function for multi-class classification. It provides strong, well-scaled gradients that facilitate stable learning.
- **MSE** is generally better suited for regression. In classification, it assumes errors are normally distributed and can cause vanishing gradients for saturated incorrect predictions.
- **MAE** is unsuitable for this classification task due to its constant gradient magnitude, which causes training instability and poor convergence.